

1. Define CMyString class with demands as follows:

- 1.1 The class, CMyString, can save any string which means an arbitrary length;
- 1.2 Overload appropriate operators to operate instance of CMyString in the main;
- 1.3 Define Find() member function to locate the first position of searching a character or a string;
- 1.4 Define Mid() member function to get substring from one string;
- 1.5 Define type operator and other member functions if you need.
- 1.6 Define any member functions if you need.
- 1.6 The client can use CMyString you defined like as in the main:

```
void main()
{
    CMyString s1("BIT"), s2;

    s2 = "love";
    s2[0] = 'L'
    cout << "I " + s2 + " " << s1 << endl;    // The output is:  I Love BIT

    int pos = s1.Find('T');    // pos is 2, or -1 if 'T' is not found.
    pos = s1.Find("IT");    // pos is 1, or -1 if "IT" is not found.

    int startPos = 3, len = 4;
    s2 = "Welcome";

    // In Mid, startPos represents starting position, len represents the length of substring.
    CMyString myStr = s2.Mid(startPos, len);    // myStr is "come"
}
```

NOTE:

1. The type such as string or template in C++, and string library functions in C and C++ are both FORBIDDEN in your program.
2. The only types you can use is building-type, such as char, int, double, bool and corresponding pointer.

2. Define CInternetURL class to save a URL string like as "www.bit.edu.cn", and finish the demands as follows:

2.1 Only use CMyString you just finished before to save URL string;

2.2 Define member function getDomain() to get domain from URL.

e.g. we can get "www.bit.edu.cn" from "https://jwc.bit.edu.cn/index.htm"

2.3 Define member function getDomainCountry() to get country name in the domain from URL, such as "us", "uk", etc.

e.g. we can get "cn" from "https://jwc.bit.edu.cn/index.htm";

2.4 Define member function getDomainType() to get domain type from URL, such as "gov", "com", "org", and so on.

e.g. we can get "edu" from "https://jwc.bit.edu.cn/index.htm";

2.5 Define member function getHomePage() to get home page from URL, such as "index.htm", "default.htm", and so on.

e.g. we can get "index" from "https://jwc.bit.edu.cn/index.htm";

2.6 The string you CAN ONLY use is CMyString in the main:

```
int main()
{
    CInternetURL URL("https://jwc.bit.edu.cn/index.htm");

    cout << URL.GetDomain();           // The result is:  www.bit.edu.cn
    cout << URL.GetDomainCountry();    // The result is:  cn
    cout << URL.GetDomainType();       // The result is:  edu
    cout << URL.GetHomePage();         // The result is:  index.htm
    return 0;
}
```

3. 银行个人账户分为贷记卡账户和借记卡账户，保存有账号、密码、余额等信息。贷记卡账户资金通过个人信用额度进行消费、取现、还款等功能。借记卡账户资金的使用通过入账、消费、取现等操作来完成。

请利用面向对象继承机制反映上述内容，并满足 `main()` 函数的调用。

```
int main()
{
    CCreditAccount credit(1061, "123456", 500); // 借记卡：账号、密码（字符串）、余额
    CDebitAccount debit(9032, "654321", 3000); // 贷记卡：账号、密码（字符串）、
                                                // 透支额度

    credit.Withdraw(200); // 取现
    cout << credit << endl; // 账户余额：300

    debit.Withdraw(200); // 取现，费率：1%，最低 2 元 / 笔
    debit.Spending(200); // 消费
    cout << debit << endl; // 账户余额：2800

    return 0;
}
```